



DPUMultiplexer

1. Inputs:

In[1...10]

Multiplicator[1...10]

Divisor[1...10]

Offset[1...10]

Switch

Next

Previous

Inputs, which can be switched to the output

Amplification factor for input

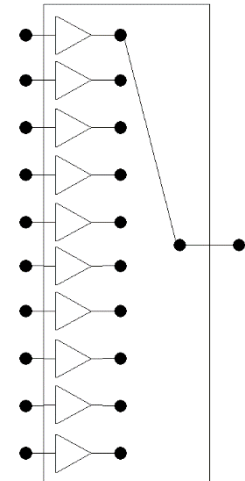
Decrease of input

Offset alignment of input

Switch of input channel

At 0→1 transition: Switch to the next input channel (*Switch* is increased by 1)

At 0→1 transition: Switch back to previous input channel (*Switch* is decreased by 1)



2. Dynamic parameters:

OutMin

OutMax

Default

NumValid

Cycle

Minimum limit of output. Default: -1e300

Maximum limit of output. Default: 1e300

Default value for output. Default: 0.0

Number of valid inputs. Default: 10

Determines what should happen in case an overflow (by *Next*) or an underflow (by *Previous*) happens. It determines whether to continue on the other end of inputs. Default: 0

3. Outputs:

Out

Output which is switched to one of the inputs.

4. Functional principle:

To the output of DPUMultiplexer one of the inputs can be switched dependant of the value of input *Switch*. If none of the inputs is switched to the output (*Switch*=0 or > 10), *Out* = *Default* is set. The input values are converted according to the following formula:

$$Out = \frac{In_{Switch} \cdot Multiplicator_{Switch}}{Divisor_{Switch}} + Offset_{Switch}$$

5. Example:

The following is an example script for using DPUMultiplexer:

```
DPUKeyboardState keyboard
{
    Computer = {LOCALHOST};
    Index = 1;
};
```



```
DPUMultiplexer multiplexer
{
    Computer = {LOCALHOST};

    Index = 1;
    In1 = 50;
    Multiplicator1 = 2;
    In2 = 50;
    Multiplicator2 = 4;
    In3 = 50;
    Multiplicator3 = 6;

    Switch = 1;
};
Connections =
{
    keyboard.D -> multiplexer.Next,
    keyboard.A -> multiplexer.Previous
};
```

In this example, three inputs are defined (In1, In2, In3). The inputs are then multiplied by two, four and six, respectively. In1 is set as the Input to be connected to the output initially. (Switch = 1) When “D” is pressed on the keyboard, the next input is selected to be displayed. When “A” is pressed on the keyboard, the previous input is selected to be displayed.